

## PROPOSED ML WORKFLOW

The proposed machine learning workflow for the HealthConnect Clinic Appointment No-Show Prediction System will provide a structured process for transforming raw appointment data into a predictive model. The workflow will focus on data quality, feature preparation, model development, and evaluation before any future deployment.

### 1. Data Ingestion

Load the HealthConnect appointment dataset into the development environment and review the accompanying Data Dictionary to understand the meaning and intended use of each variable.

### 2. Data Validation and Inspection

Assess the dataset for:

- Number of records and features
- Data types
- Missing values
- Duplicate records
- Invalid or inconsistent values
- Target-variable distribution
- Potential data-quality issues

The current dataset contains 5,000 records and 18 variables.

### 3. Data Cleaning

Handle identified data-quality issues appropriately.

The current missing-value findings include:

reminder\_channel: 1,366 missing values

distance\_to\_clinic\_km: 90 missing values

waiting\_time\_minutes: 60 missing values

The reminder\_channel missing values have been identified as corresponding to appointments where no reminder was sent, so they can be represented as "No Reminder" rather than being treated as ordinary missing observations.

### 4. Feature Engineering

Transform existing variables into useful machine-learning features where justified.

Potential activities include:

Extracting useful information from date variables

Reviewing age and age\_group for redundancy

Reviewing appointment\_date and appointment\_day

Creating meaningful lead-time groups where appropriate

Preparing numerical and categorical variables for modelling

## **5. Target Definition**

Define the prediction target using appointment\_outcome.

The primary business focus is No-Show prediction. The treatment of Cancelled appointments will be finalized before model training.

## **6. Feature Selection**

Identify features that are relevant to the prediction task while removing:

Unique identifiers such as appointment\_id

patient\_id as a predictive feature

The target variable from the input features

Redundant variables

Features that could introduce data leakage

Special attention will be given to whether variables such as waiting\_time\_minutes are available at the time the prediction is made.

## **7. Data Splitting**

Separate the prepared data into training and testing datasets.

Because the same patient may have multiple appointments, the splitting strategy should consider patient-level grouping to reduce the risk of information from the same patient appearing in both training and testing data.

## **8. Preprocessing Pipeline**

Build a reproducible preprocessing pipeline that performs the required transformations, including:

Numerical missing-value imputation

Categorical missing-value handling

Categorical encoding

Numerical scaling where required

Preprocessing parameters should be learned from the training data to reduce the risk of data leakage.

## **9. Model Development**

Train and compare proposed classification algorithms.

The initial candidates are:

Logistic Regression — baseline model

Random Forest — nonlinear tree-based model

XGBoost — gradient-boosting model

At this stage, these are proposed models, not completed model results.

## **10. Model Evaluation**

Evaluate model performance using appropriate classification metrics:

Accuracy

Precision

Recall

F1-score

ROC-AUC

Confusion matrix

Particular attention will be given to Recall for the No-Show class, because identifying appointments that are genuinely at risk of being missed is important to the business objective.

## **11. Model Comparison and Selection**

Compare the candidate models based on their predictive performance and practical usefulness.

The final model should be selected based on a combination of:

Predictive performance

No-Show recall

Precision

F1-score

ROC-AUC

Interpretability

Operational suitability

## 12. Risk Prediction

The selected model will produce a probability or risk score representing the likelihood that an appointment will result in a No-Show.

The prediction can subsequently be translated into appropriate risk categories for operational use.

## 13. Future Deployment

After successful validation, the model could eventually be integrated into HealthConnect's appointment-management workflow.

This would allow the clinic to identify potentially high-risk appointments early and consider appropriate patient-engagement strategies.

### WORKFLOW SUMMARY

